

Executive Business Report

End-to-End Sales Forecasting & Demand Intelligence System

Executive Summary

This project developed an intelligent sales forecasting and demand analysis system to support inventory planning and business decision-making. Historical sales data was analyzed to identify trends, seasonality, demand patterns, anomalies, and product performance. Three forecasting models—SARIMA, Prophet, and XGBoost—were evaluated, with XGBoost providing the highest forecasting accuracy based on MAE, RMSE, and MAPE. The system also detects unusual sales behaviour, segments products into demand groups, and provides an interactive Streamlit dashboard that enables managers to explore forecasts, anomalies, and product demand insights for better stocking decisions.

Key Findings

- Sales exhibit a clear upward trend with recurring seasonal patterns across the four-year period.
 - Technology generated the highest historical revenue, while Furniture is forecasted to experience the strongest upcoming growth.
 - XGBoost achieved the best forecasting performance among the evaluated models.
 - The West region is expected to show stronger future demand than the East region.
 - Product demand varies significantly across sub-categories, making segmentation valuable for inventory optimization.
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Three-Month Sales Forecast

The forecasting model predicts continued business growth over the next three months.

- Furniture sales are expected to increase significantly.
- The West region is projected to maintain strong demand.
- Office Supplies are expected to remain relatively stable.
- Technology demand should be monitored because the model predicts a slight decline compared to other segments.

Forecasts should be reviewed regularly as new sales data becomes available.

Top Anomalies Detected

Anomaly 1

A significant increase in weekly sales was detected during the holiday shopping season.

Likely Cause: Promotional campaigns or festive shopping.

Anomaly 2

An unusually low sales week was identified shortly after the holiday period.

Likely Cause: Reduced customer demand following seasonal purchasing.

Anomaly 3

A sudden sales spike occurred outside the normal seasonal trend.

Likely Cause: Special discounts, flash sales, or bulk customer orders.

Product Demand Segmentation

Four demand clusters were identified using K-Means clustering:

Segment	Recommended Strategy
High Volume, Stable Demand	Maintain consistent inventory with automated replenishment.
Growing Demand	Increase safety stock and monitor demand frequently.
Low Volume, High Volatility	Keep limited inventory and replenish based on actual demand.
Declining Demand	Reduce stock levels and avoid over-ordering.

Business Recommendations

1. Increase inventory allocation for Furniture products and the West region due to strong forecasted demand.
 2. Monitor Technology inventory carefully to avoid overstocking as the forecast indicates slower growth.
 3. Use anomaly detection to investigate unusual sales periods and improve promotional planning and inventory management.
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Risk / Limitation

The forecasting models rely solely on historical sales data. External factors such as economic conditions, competitor actions, supply chain disruptions, or unexpected market events are not included in the model and may affect forecast accuracy.